

Letters

Robust GNSS/5G Positioning in LOS/NLOS Environments: A Stein Variational Inference Framework for Particle Filter

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Abstract—High-accuracy positioning in fifth generation (5G) industrial networks is severely challenged by nonlinear-of-sight (NLOS) propagation, which induces particle degeneracy in conventional particle filters. This letter proposes a robust Stein variational particle filter (RSVPF) that addresses this limitation. By reformulating the problem as a variational inference task, the RSVPF deterministically transports particles to approximate the optimal proposal distribution via a Stein variational gradient flow. The method operates on a hybrid state-space, jointly estimating kinematic state and sight conditions, and is complemented by a projection onto a physically feasible subspace. The proposed framework is shown to mitigate particle impoverishment, enhance robustness against NLOS bias, and provide accurate, stable posterior estimation for integrated global navigation satellite system (GNSS)/5G positioning.

Index Terms—Fifth generation (5G) positioning, Bayesian filtering, global navigation satellite system (GNSS), nonlinear-of-sight (NLOS) mitigation, particle filter (PF), stein variational inference.

I. INTRODUCTION

THE convergence of fifth generation (5G) networks and the global navigation satellite system (GNSS) is critical for high-accuracy positioning in applications, such as industrial automation and autonomous vehicles [1], [2]. However, performance in urban environments is severely degraded by nonlinear-of-sight (NLOS) propagation [3]. The resulting large, non-Gaussian measurement biases violate the assumptions of standard algorithms, such as the Kalman filter, posing a significant challenge [4].

To contend with such nonlinear and non-Gaussian systems, particle filters (PFs) have emerged as a powerful and widely adopted solution [5]. By representing the posterior density with a set of weighted random samples, PFs can, in theory, approximate any probability distribution. However, the practical efficacy of a PF is critically dependent on its proposal distribution. The

standard approach, which uses the transition prior for proposal generation, is notoriously inefficient when the likelihood is informative but concentrated in a region of low prior probability—a scenario emblematic of NLOS conditions. This mismatch leads to the well-documented problem of particle degeneracy, where a few particles acquire overwhelmingly large weights, causing a collapse of the Monte Carlo approximation and a failure of the filter [6].

While various techniques have been proposed for NLOS mitigation, including cooperative localization [7], GNSS/5G positioning, and map-based methods [8], they often address the problem at a systemic level. The fundamental weakness of the underlying filtering mechanism in the presence of severe NLOS bias remains a significant open challenge. Existing PF enhancements, such as auxiliary or extended PFs, offer only partial remedies and may struggle when the posterior becomes highly multimodal or complex.

In this letter, we address this foundational challenge by proposing a novel filtering paradigm: the robust stein variational particle filter (RSVPF). By casting the problem within a variational inference framework, the RSVPF abandons random sampling from the prior. Instead, it deterministically transports particles to approximate the optimal proposal distribution via a Stein variational gradient flow, guided by the functional gradient of the Kullback–Leibler (KL) divergence.

The core contribution of this letter is a *novel filtering paradigm*, the RSVPF. This new framework advances the subject area by overcoming the fundamental *particle degeneracy* problem: a chronic issue that cripples conventional PFs when the transition prior mismatches the highly informative, non-Gaussian NLOS likelihood. Our specific contributions, which synergistically deliver this new framework, are as follows.

- 1) We introduce a variational inference framework for hybrid-state GNSS/5G navigation that jointly estimates the continuous kinematic state and the discrete sight conditions.
- 2) We develop a novel proposal mechanism based on stein variational importance sampling (SV-IS) that *deterministically* generates high-quality particles, thereby mitigating the sample impoverishment that plagues random sampling methods.
- 3) We synergistically combine the “soft” optimization of the variational gradient flow with a “hard” projection onto a

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physically feasible subspace, enhancing filter robustness and integrity.

The rest of this letter is organized as follows. Section II presents the theoretical foundation and detailed mathematical formulation of the proposed RSVPF. Section III showcases the superior performance of our method through simulation results. Finally, Section IV concludes this letter.

II. PROPOSED RSVPF: A VARIATIONAL INFERENCE FRAMEWORK

We formulate the problem of integrated navigation as the recursive estimation of a sequence of posterior probability measures, $(\pi_t)_{t \geq 0}$, defined over a hybrid, continuous-discrete state space. The proposed RSVPF is a principled computational methodology to approximate these measures. The core innovation lies in constructing the particle proposal mechanism as a variational flow that minimizes the KL divergence to the analytically intractable optimal importance density. This section delineates the mathematical foundations of this approach.

A. Probabilistic Model Formulation on a Hybrid State Space

Let the state space be the product space $\mathcal{Z} \triangleq \mathbb{R}^d \times \{0, 1\}^{\mathcal{M}}$, where $d = 2$ is the dimension of the kinematic state and $\mathcal{M}$ denotes the number of base stations (BSs). An element $\mathbf{z}_t = (\mathbf{x}_t, \mathbf{s}_t) \in \mathcal{Z}$ comprises the continuous position vector $\mathbf{x}_t$ and the discrete binary vector of sight conditions $\mathbf{s}_t$. The probabilistic structure of the system is defined by a state evolution model and a measurement process.

1) *State Evolution Model*: The evolution of the state is governed by a Markov kernel $K_t : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{Z})$, where $\mathcal{P}(\mathcal{Z})$ is the space of probability measures on $\mathcal{Z}$. We assume that the kinematic and sight-condition components evolve independently, conditioned on the past state, yielding a transition density of the form $p(\mathbf{z}_t | \mathbf{z}_{t-1}) = p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_{t-1})p(\mathbf{s}_t | \mathbf{s}_{t-1})$. Taking into account the presence of an input control with sampling period Δt , the linear model for the state transition is

$$\mathbf{x}_t = \mathbf{F}\mathbf{x}_{t-1} + \mathbf{B}\mathbf{u}_{t-1} + \boldsymbol{\omega}_{t-1}, \quad \boldsymbol{\omega}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad (1)$$

where the state transition matrix $\mathbf{F} \in \mathbb{R}^{2 \times 2}$ is a 2×2 identity matrix. The control input $\mathbf{u}_{t-1}$ is the velocity $\mathbf{v}_G(t-1)$ provided by the GNSS module. The control-input matrix $\mathbf{B} \in \mathbb{R}^{2 \times 2}$ is given by

$$\mathbf{B} = \begin{bmatrix} \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}. \quad (2)$$

The process noise $\boldsymbol{\omega}_{t-1}$ is assumed to be zero-mean Gaussian noise, where the covariance matrix $\mathbf{Q}$ describes the uncertainty of the state.

2) *5G Measurement*: The 5G network provides the measurement observations $\mathbf{y}_t$ (e.g., ranging data) used for the filter's update step. The channel impulse response (CIR) is simulated by ray-tracing (RT) [9]. A generic model for the simulated CIR

is modeled as

$$h(\tau) = \sum_{l=0}^{L-1} \alpha_l \delta(\tau - \tau_l) \quad (3)$$

where L is the number of multipath components, α_l and τ_l are the relative attenuation and delay components, respectively, of the l th path with respect to the first path.

The measurement process is characterized by a likelihood function $g_t : \mathcal{Z} \rightarrow \mathbb{R}^+$, $g_t(\mathbf{z}_t) \triangleq p(\mathbf{y}_t | \mathbf{z}_t)$. The observation $\mathbf{y}_t \in \mathbb{R}^{\mathcal{M}}$ is conditionally independent given the state, hence $g_t(\mathbf{z}_t) = \prod_{i=1}^{\mathcal{M}} p(y_t^i | \mathbf{x}_t, s_t^i)$. For the i th BS, the 5G time-of-arrival (TOA) measurement model is

$$y_t^i = \|\mathbf{x}_t - \mathbf{p}_{\text{BS},i}\| + v_{\text{los}}^i + (1 - s_t^i)v_{\text{nlos}}^i \quad (4)$$

where $\mathbf{p}_{\text{BS},i}$ is the known position vector of the i th BS, $v_{\text{los}}^i \sim \mathcal{N}(0, \sigma_{\text{los},i}^2)$ and v_{nlos}^i is a random variable with a positive support, modeling the line-of-sight (LOS) and NLOS bias, respectively.

B. RSVPF Methodology

The RSVPF approximates the posterior measure $\pi_t(d\mathbf{z}_t) \triangleq p(\mathbf{z}_t | \mathcal{D}_t)d\mathbf{z}_t$, where $\mathcal{D}_t$ is the filtration generated by the observation and control history.

1) *Bayesian Recurrence via Integral Operators*: The evolution of the posterior measure is governed by a two-stage recurrence. First, the prediction operator $\mathcal{P}_t$ evolves the previous posterior π_{t-1} to the predictive measure $\pi_{t|t-1}$ via the Chapman-Kolmogorov equation

$$\pi_{t|t-1}(A) = (\mathcal{P}_t \pi_{t-1})(A) = \int_{\mathcal{Z}} K_t(\mathbf{z}_{t-1}, A) \pi_{t-1}(d\mathbf{z}_{t-1}). \quad (5)$$

Second, the update operator $\mathcal{U}_t$, an application of Bayes' theorem, incorporates the new information from measurement $\mathbf{y}_t$ via the likelihood g_t

$$\pi_t(d\mathbf{z}_t) = (\mathcal{U}_t \pi_{t|t-1})(d\mathbf{z}_t) = \frac{g_t(\mathbf{z}_t) \pi_{t|t-1}(d\mathbf{z}_t)}{Z_t} \quad (6)$$

where $Z_t \triangleq \int_{\mathcal{Z}} g_t(\mathbf{z}') \pi_{t|t-1}(d\mathbf{z}') = p(\mathbf{y}_t | \mathcal{D}_{t-1})$ is the marginal likelihood or evidence, serving as the normalization constant.

2) *Variational Approximation of the Optimal Proposal*: The performance of any importance sampler is contingent upon the choice of the proposal measure q_t .

Proposition 1: The proposal measure $q_t(\mathbf{z}_t | \mathbf{z}_{t-1})$ that minimizes the conditional variance of the importance weights, $\text{Var}(w_t | \mathbf{z}_{t-1})$, is the posterior predictive measure $p(\mathbf{z}_t | \mathbf{z}_{t-1}, \mathbf{y}_t)$.

Proof: The importance weight is $w_t \propto g_t(\mathbf{z}_t) p(\mathbf{z}_t | \mathbf{z}_{t-1}) / q_t(\mathbf{z}_t | \mathbf{z}_{t-1})$. Selecting $q_t = q_{\text{opt}} \propto g_t(\mathbf{z}_t) p(\mathbf{z}_t | \mathbf{z}_{t-1})$ renders the weight independent of the current state $\mathbf{z}_t$, making it proportional to the local evidence $p(\mathbf{y}_t | \mathbf{z}_{t-1})$, thereby minimizing variance. ■

Direct sampling from this optimal density, q_{opt} , is intractable. We, therefore, construct a measure $q_{\text{SV-IS}}$ that approximates it through a variational approach. This is framed as a functional gradient descent on the KL divergence $\text{KL}(q || q_{\text{opt}})$ within a reproducing Kernel Hilbert space $\mathcal{H}$ associated with a positive

definite kernel $\mathcal{K}(\cdot, \cdot)$. The velocity field $\phi \in \mathcal{H}$ that maximally decreases the KL divergence is given by the Stein operator

$$\phi^*(\cdot) = \mathbb{E}_{\mathbf{x} \sim q} [\mathcal{K}(\mathbf{x}, \cdot) \nabla_{\mathbf{x}} \log q_{\text{opt}}(\mathbf{x}) + \nabla_{\mathbf{x}} \mathcal{K}(\mathbf{x}, \cdot)] \quad (7)$$

where q is the current empirical distribution of the particles. The required log-gradient of the target density is analytic and given by

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log q_{\text{opt}}(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}, \mathbf{s}_t^{(j)}, \mathbf{y}_t, \mathbf{u}_{t-1}) = \\ \nabla_{\mathbf{x}_t} \log g_t(\mathbf{x}_t, \mathbf{s}_t^{(j)}) + \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}, \mathbf{u}_{t-1}). \end{aligned} \quad (8)$$

Iteratively applying this gradient flow to a set of particles generates an empirical measure that converges to an approximation of q_{opt} .

3) LOS/NLOS Projection onto the Feasible Subspace: The variational gradient flow of SV-IS is highly effective at shaping the empirical measure to capture the modes of the target density. However, this gradient-based “soft” optimization may not strictly enforce hard constraints derived from physical principles, particularly those that define nondifferentiable LOS/NLOS boundaries. To address this, we introduce a projection step that serves as a theoretical complement, injecting infeasible domain knowledge into the posterior approximation.

Let $\hat{\pi}_{\text{SV-IS}}^N(d\mathbf{x}) = \sum_{j=1}^N w_{\text{SV-IS}}^{(j)} \delta_{\mathbf{x}_t^{(j)}}(d\mathbf{x})$ be the weighted empirical measure after the SV-IS transport. We define a feasible subspace $D^{(j)} \subset \mathbb{R}^d$ for each particle, conditioned on its sampled sight state $s_{t,i}^{(j)} \in \{0, 1\}^M$

$$D^{(j)} = \left(\bigcap_{i: s_{t,i}^{(j)}=1} D_{\text{LOS}}^{(i)} \right) \cap \left(\bigcap_{i: s_{t,i}^{(j)}=0} D_{\text{NLOS}}^{(i)} \right) \quad (9)$$

where $D_{\text{LOS}}^{(i)} \triangleq \{\mathbf{x} \mid \|\mathbf{y}_t^i - \|\mathbf{x} - \mathbf{p}_{\text{BS},i}\|\| \leq \gamma \sigma_{\text{LOS},i}\}$ and $D_{\text{NLOS}}^{(i)} \triangleq \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{p}_{\text{BS},i}\| \leq y_t^i\}$. γ is a confidence coefficient (typically 2 or 3) that scales the LOS noise deviation to define the admissible residual tolerance. The projection is then realized by multiplying the particle weights with the indicator function $\mathbb{I}_{D^{(j)}}(\cdot)$ of this feasible set. This operation effectively discards particles that violate the physical constraints by assigning them zero probability mass. The weight update is thus

$$w_{\text{proj}}^{(j)} = w_{\text{SV-IS}}^{(j)} \cdot \mathbb{I}_{D^{(j)}}(\mathbf{x}_t^{(j)}) \quad (10)$$

where

$$\mathbb{I}_D(x) \triangleq \begin{cases} 1, & \text{if } x \in D \\ 0, & \text{if } x \notin D. \end{cases} \quad (11)$$

4) Importance Weighting as a Radon–Nikodym Derivative: The importance weight is formally the Radon–Nikodym derivative of the target posterior measure with respect to the proposal measure, $w_t = d\pi_t/dq_t$. For a particle $(\mathbf{z}_{t-1}^{(j)}, w_{t-1}^{(j)})$, the updated unnormalized weight $\tilde{w}_t^{(j)}$ is derived as follows:

$$\tilde{w}_t^{(j)} \propto w_{t-1}^{(j)} \frac{p(\mathbf{y}_t | \mathbf{z}_t^{(j)}) p(\mathbf{z}_t^{(j)} | \mathbf{z}_{t-1}^{(j)})}{q_t(\mathbf{z}_t^{(j)} | \mathbf{z}_{t-1}^{(j)}, \mathbf{y}_t)}$$

$$\begin{aligned} &= w_{t-1}^{(j)} \frac{\overbrace{p(\mathbf{y}_t | \mathbf{x}_t^{(j)}, \mathbf{s}_t^{(j)})}^{\text{Likelihood}} \overbrace{p(\mathbf{x}_t^{(j)} | \mathbf{x}_{t-1}^{(j)}, \mathbf{u}_{t-1}) p(\mathbf{s}_t^{(j)} | \mathbf{s}_{t-1}^{(j)})}^{\text{Prior Evolution}}}{\underbrace{q_{\text{SV-IS}}(\mathbf{x}_t^{(j)} | \dots)}_{\text{Kinematic Proposal}} \underbrace{p(\mathbf{s}_t^{(j)} | \mathbf{s}_{t-1}^{(j)})}_{\text{Sight State Proposal}}}. \end{aligned} \quad (12)$$

The density for the sight state proposal $p(\mathbf{s}_t^{(j)} | \mathbf{s}_{t-1}^{(j)})$ cancels, yielding the simplified expression

$$\tilde{w}_t^{(j)} \propto w_{\text{proj}}^{(j)} \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(j)}, \mathbf{s}_t^{(j)}) p(\mathbf{x}_t^{(j)} | \mathbf{x}_{t-1}^{(j)}, \mathbf{u}_{t-1})}{q_{\text{SV-IS}}(\mathbf{x}_t^{(j)} | \mathbf{z}_{t-1}^{(j)}, \mathbf{y}_t)}. \quad (13)$$

The variational procedure is explicitly designed to force the proposal density $q_{\text{SV-IS}}$ to closely mirror the numerator. The proximity of the numerator and denominator, enforced by the KL minimization, ensures that this Radon–Nikodym derivative remains in the vicinity of unity, thereby guaranteeing the stability of the Monte Carlo approximation of the posterior measure π_t .

Conventional PFs fail in NLOS conditions due to *particle degeneracy*; the large, non-Gaussian bias creates a strong likelihood in a region far from the transition prior, causing a few particles to receive all the weight. Our RSVPF framework counteracts this through the following *synergistic two-pronged approach*.

- 1) *Deterministic transport:* Instead of random sampling, we use a *Stein variational gradient flow* (detailed in Section II-B-2). This is an *optimization* process that deterministically transports all particles from their prior locations to approximate the true, complex posterior (which is shaped by the NLOS likelihood). This fundamentally avoids the degeneracy caused by the prior-likelihood mismatch.
- 2) *Hybrid-state projection:* We explicitly model the LOS/NLOS sight condition as part of the state. This allows us to combine the “soft” optimization of SV-IS with a “hard” *geometric projection* (detailed in Section II-B-3). This projection enforces infeasible physical constraints, effectively assigning zero weight to particles that are inconsistent with their claimed sight-state.

This combination of gradient-based optimization and hard physical constraints is the key to the filter’s robustness against severe NLOS bias.

5) Complexity Analysis: A formal analysis of the computational complexity is warranted for assessing practicality. The complexity of a standard PF [and variants, such as the auxiliary PF (APF)] [10] is typically $O(N)$ or $O(N \log N)$, dominated by the resampling stage. The proposed RSVPF, however, has its computational bottleneck in the iterative Stein variational gradient flow. The calculation of the Stein operator (5) for each of the N particles requires a sum over all other $N - 1$ particles, resulting in an $O(N^2)$ complexity for a single gradient step. As this is repeated for L iterations, the proposal stage has a total complexity of $O(L \cdot N^2)$. This higher computational cost is the necessary and deliberate tradeoff for avoiding the particle degeneracy that plagues sampling-based filters in this

Algorithm 1: Proposed RSVPF Framework.

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1: Input: Particle set  $\{\mathbf{z}_{t-1}^{(j)}, \mathbf{w}_{t-1}^{(j)}\}_{j=1}^N$  representing  $\pi_{t-1}$ ,
   new measurement  $\mathbf{y}_t$ , GNSS velocity control  $\mathbf{u}_{t-1}$ .
2: Output: Updated particle set  $\{\mathbf{z}_t^{(j)}, \mathbf{w}_t^{(j)}\}_{j=1}^N$ 
   representing  $\pi_t$ .
3: for  $t = 1, 2, 3, \dots$  do do
4:   for  $j = 1$  to  $N$  do do
5:     1. Prediction (Particle Initialization)
6:     Sample sight state:  $\mathbf{s}_t^{(j)} \sim p(\mathbf{s}_t | \mathbf{s}_{t-1}^{(j)})$ 
7:     Sample initial kinematic state:
        $\hat{\mathbf{x}}_t^{(j)} \sim p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}, \mathbf{u}_{t-1})$  according to (1).
9:     2. Variational Transport (SV-IS)
10:    Transport  $\hat{\mathbf{x}}_t^{(j)}$  to  $\mathbf{x}_t^{(j)}$  via  $L$  Stein gradient steps
       according to equations (5)-(6).
11:    Set final particle state  $\mathbf{z}_t^{(j)} = (\mathbf{x}_t^{(j)}, \mathbf{s}_t^{(j)})$ 
13:    3. Projection and Weight Update
14:    Define feasible subspace  $D^{(j)}$  using (7) based on
        $\mathbf{s}_t^{(j)}$ .
15:    Calculate projection factor  $w_{proj}^{(j)} = \mathbb{I}_{D^{(j)}}(\mathbf{x}_t^{(j)})$ 
       using (8).
16:    Update unnormalized weight  $\tilde{w}_t^{(j)}$  according to
       (11), incorporating  $w_{t-1}^{(j)}$  and  $w_{proj}^{(j)}$ .
17:   end for
19:   4. Normalization
20:   Calculate total weight:  $W_t = \sum_{i=1}^N \tilde{w}_t^{(i)}$ 
21:   for  $j = 1$  to  $N$  do do
22:      $w_t^{(j)} = \tilde{w}_t^{(j)} / W_t$ 
23:   end for
24: end for

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application. While the $O(N)$ PF fails, the $O(L \cdot N^2)$ RSVPF succeeds by deterministically transporting a small, fixed particle set ($N = 200$) to the high-probability posterior regions.

III. NUMERICAL VALIDATION AND PERFORMANCE ANALYSIS

In this section, we conduct a comprehensive numerical analysis to validate the proposed theoretical framework and quantify the performance gains of the RSVPF for GNSS/5G integrated navigation. The simulations are contextualized within a canonical 5G ultra-dense network scenario, meticulously modeled to reflect realistic propagation challenges.

A. Simulation Setup and Data Generation

We consider a trajectory within an industrial urban micro environment populated with 5G BSs. The ground-truth trajectory and GNSS control inputs are generated based on a constant velocity model ($v = 1\text{m/s}$), perturbed by zero-mean Gaussian noise consistent with a commercial-grade receiver (e.g., Garmin GPS 18x-5 Hz), characterized by a velocity error of 0.1knots [11]. More detailed 5G environment settings can be found in our previous studies [12], [13], [14], [15], [16].

The 5G ranging measurements are synthesized to encapsulate both *LOS* and *NLOS* phenomena.

- 1) *LOS noise*: The variance of the zero-mean Gaussian noise component, $\sigma_{v_{\text{los},i}}^2$, is derived from the Cramér–Rao lower bound for TOA estimation in orthogonal frequency-division multiplexing (OFDM) systems [17]

$$\sigma_{v_{\text{los},i}}^2 = \frac{c^2 T_s^2}{8\pi^2 \cdot \text{SNR}_i \cdot \sum_{n \in \mathcal{N}_{\text{PRS}}} p_n^2 \cdot n^2} \quad (14)$$

where c is the speed of light, T_s is the OFDM symbol duration, SNR_i is the signal-to-noise ratio (SNR), $\mathcal{N}_{\text{PRS}}$ is the set of subcarrier indices for the positioning reference signal (PRS), and p_n is the power weighting of the n th subcarrier.

- 2) *NLOS bias*: The positive bias term, v_{nlos}^i , is generated based on realistic CIRs simulated via ray tracing (RT). Following the model in [9], the bias is a function of the geometric properties of the CIR

$$v_{\text{nlos}}^i = \frac{c(\sin(\frac{\pi}{2N_a}))^3}{4\pi F_{\text{sc}} \cos(\frac{\pi}{2N_a})} (\chi_1 + \chi_2) \quad (15)$$

where F_{sc} is the subcarrier spacing (with $T_s = 1/F_{\text{sc}}$), N_a is the number of active subcarriers, and the terms χ_1 and χ_2 represent bias components derived from the geometric properties of the CIR's multipath profile.

B. Performance Evaluation and Analysis

We evaluate the performance of the proposed RSVPF against four benchmark algorithms: the GNSS-only solution, the standard PF, interacting multiple model extended Kalman filter (IMM-EKF) [18], and APF. We evaluate the performance of the proposed RSVPF against three benchmark algorithms, selected to provide a comprehensive comparison. The *standard PF* is included as the foundational benchmark to demonstrate the critical problem of particle degeneracy that our method is designed to solve. The *IMM-EKF* [16] is used as a strong, model-based baseline that explicitly handles switching dynamics, but is limited by Gaussian assumptions in non-Gaussian NLOS conditions. Finally, the *GNSS-only* solution is used to quantify the overall gain from 5G fusion. While other advanced PFs, such as the APF, exist, they are enhancements to the underlying sequential importance sampling with resampling (SIS-R) framework. Our RSVPF, in contrast, is a different paradigm based on variational inference and deterministic particle transport, and our comparison is designed to highlight the fundamental advantages of this new approach.

The proposed RSVPF was implemented with $N = 200$ particles and $L = 5$ Stein gradient steps ($\varepsilon = 0.1$), using a Gaussian radial basis function (RBF) kernel to define the local, exponentially decaying interparticle influence based on squared Euclidean distance. To validate the robustness of the RSVPF to its hyperparameters, we performed a sensitivity analysis on the number of Stein gradient steps (L) and the RBF kernel bandwidth. We evaluated the root-mean-square error (RMSE) for $L \in \{1, 3, 5, 10\}$. The results showed that the performance was suboptimal with $L = 1$ but improved significantly at $L = 3$ and stabilized at $L = 5$. The performance gain from $L = 5$ to

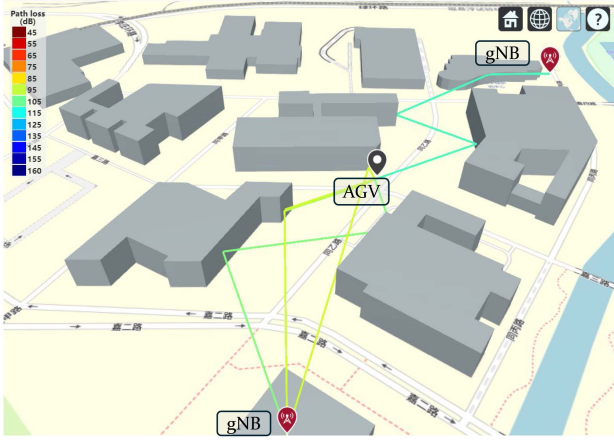


Fig. 1. Snapshot of the MATLAB RT tool showing a simulated 5G positioning scenario, where an autonomous ground vehicle (AGV) (black marker) navigates in an environment surrounded by two generation NodeBs (gNBs) (red markers).

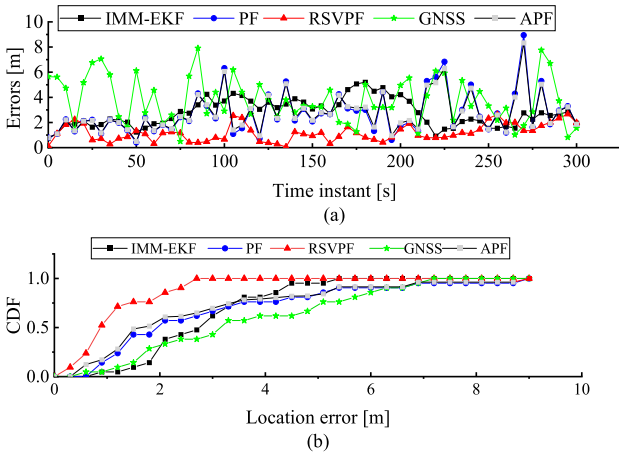


Fig. 2. Localization error analysis. (a) Localization error along the path. (b) CDF of localization error.

$L = 10$ was negligible, justifying our choice of $L = 5$ as offering a strong balance between accuracy and computational cost. Furthermore, the filter's performance remained stable across a reasonable range of kernel bandwidths, indicating that the RSVPF does not require impractical, fine-grained parameter tuning to achieve the reported robust performance.

To ensure the simulation accurately represents a challenging 5G positioning scenario, system-level performance is evaluated within a high-fidelity simulation platform built on MATLAB RT tool [19]. This platform models the propagation of 3GPP Release 16-compliant PRS and sounding reference signals within detailed 3-D urban environments. It captures building-induced multipath effects with a configurable number of reflections, providing a realistic and reproducible framework for analysis. The *site viewer* feature enables visualization of these scenarios, an example of which is depicted in Fig. 1, featuring an (blue) navigating between two gNodeBs (BSs and red).

Fig. 2(a) and 2(b) illustrates the localization error along the trajectory and its cumulative distribution function (CDF), respectively. It is evident that the proposed RSVPF consistently

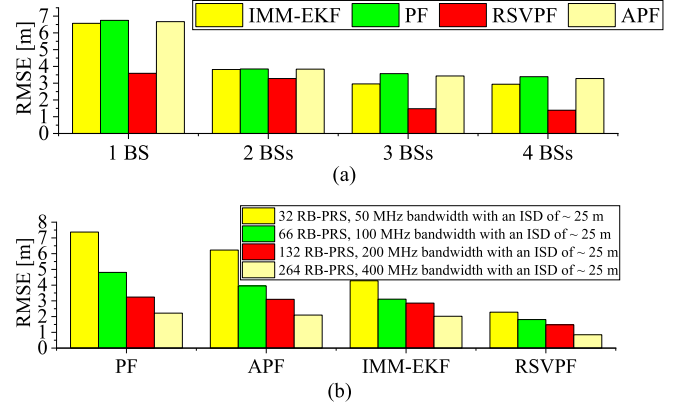


Fig. 3. RMSE analysis. (a) RMSE for all positioning methods with different number of BSs. (b) RMSE for all positioning methods under different simulation numerologies with an intersite distance of 25 m.

and significantly outperforms all baselines, achieving stable meter-level accuracy. This superior performance is directly attributed to the variational proposal mechanism. By deterministically transporting particles to regions of high posterior probability, the RSVPF effectively avoids the particle degeneracy that plagues the standard PF, especially when confronted with informative but non-Gaussian NLOS likelihoods. The PF, relying on the transition prior, frequently fails to place particles in the region supported by the likelihood, resulting in poor accuracy. The APF shows a notable improvement over the standard PF by incorporating measurement information into the proposal step. However, as an enhanced sampling-based method, it still struggles to fully capture the complex posterior in severe NLOS conditions and is clearly outperformed by the deterministic RSVPF. The IMM-EKF performs better than the PF by explicitly modeling the switching dynamics, but its Gaussian assumptions limit its ability to capture the true, often skewed, posterior density in NLOS conditions.

Further analysis is presented in Fig. 3, which evaluates the framework's robustness from two key perspectives: robustness to geometric diversity and NLOS bias [see Fig. 3(a)], and robustness to LOS measurement noise [see Fig. 3(b)]. As shown in Fig. 3(a), the accuracy of all 5G-based methods improves with the number of available BSs. This is because more measurements provide greater geometric diversity and information to reject NLOS biases. Critically, our RSVPF shows the most significant advantage when geometry is poor (e.g., 1 or 2 BSs), demonstrating its superior robustness in challenging NLOS-dominant scenarios compared to the APF and the specialized IMM-EKF. Fig. 3(b) provides a direct analysis of robustness to *LOS measurement noise* by demonstrating the impact of signal bandwidth. As confirmed by (12), higher bandwidths (and thus a larger number of resource blocks, N_{RB}) lead to lower measurement variance. While all fusion algorithms benefit from this, the RSVPF maintains the lowest RMSE across all noise levels. This twofold analysis confirms that the proposed RSVPF is robust not only to varying levels of Gaussian noise but, more importantly, to the severe non-Gaussian biases characteristic of NLOS propagation, where other methods degrade.

IV. CONCLUSION

This letter has introduced an RSVPF, a *novel filtering paradigm* that fundamentally advances the state-of-the-art in robust GNSS/5G positioning. We have addressed the intrinsic limitations of the “old state” of filtering—which relies on either 1) random sampling-based PFs (such as APF) that suffer from particle degeneracy, or 2) model-based filters (such as IMM-EKF) that are constrained by Gaussian assumptions. By *reformulating particle filtering as a deterministic optimization problem*, our proposed approach leverages deterministic particle transport to solve the prior-likelihood mismatch. This *paradigm shift* effectively alleviates particle degeneracy and allows for the accurate approximation of the true, non-Gaussian posterior. The synergistic integration of this “soft” transport with a “hard” geometric projection further enhances resilience against severe NLOS biases. As demonstrated in Figs. 2 and 3, the RSVPF achieves superior positioning accuracy and stability where all baselines degrade, making it a promising and structurally robust solution for submeter localization in 5G-and-beyond networks.

Future work will focus on validating this framework on a real-world experimental platform. We anticipate several key implementation challenges: 1) adapting the filter to handle the highly asynchronous and multirate nature of real-world GNSS and 5G data streams; 2) addressing the “sim-to-real” gap, particularly in the stochastic modeling of NLOS bias and the noise parameters governing the geometric projection; and 3) optimizing the $O(L \cdot N^2)$ Stein gradient flow computation to achieve real-time performance on a resource-constrained embedded system. Overcoming these practical hurdles will be the focus of our continued research.

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